

Shenqiu Huang (Henry)

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EDUCATION: **Southern Methodist University** - Bobby B. Lyle School of Engineering
Bachelor of Science in Electrical and Computer Engineering
Minor in Music – Saxophone Performance
Embrey Academic Excellence Scholarship
Dallas, TX
May 2027
GPA: 3.62

RELEVANT COURSES: Introduction to Electrical and Computer Engineering, Digital Computer Logic, Microcontrollers
Embedded Systems, Design and Analysis of Signals and Systems, Computer Science

SKILLS: **Languages:** Java, ARM Assembly, Python, C++, Verilog, HTML
Software: AutoCAD, Autodesk Fusion 360, Visual Studio Code, UltiMaker Cura, Logic, Arduino
Professional: Collaboration, Research, Data Analysis, Procedure Implementation

PROJECTS:

Communication Protocol, Independent Research Dec 2023 – Jan 2024

- Conducted in-depth analysis and comparison of UART, SPI, and I2C communication protocols, delivering the final presentation and submitting the comprehensive research paper for publication.
- Published the paper in the 2023 IEEE 2nd International Conference on Electrical Engineering, Big Data, and Algorithms (EEBDA).

Bike Computer Mount, Open-Source Project Aug 2022 – Dec 2022

- Utilized Fusion 360 to create a custom Bike Computer mount, modeling and performing static simulation analysis to ensure durability and functionality.
- Completed a five-month iterative development process with four resin test prints, culminating in a durable PA12 nylon prototype successfully field-tested by cycling club members.

Cycling Balance Sensors, Independent Research Aug 2021 – Dec 2021

- Researched Cycling Balance Sensors & Statistic Analysis, using statistical knowledge to program and analyze the motion data of cyclists
- Collaborated with an Oxford University Alumni, learning advanced analytical and academic presentation skills.

EXPERIENCE:

Southern Methodist University, Dallas, TX

Research Assistant, Physics Lab - ATLAS Project

May 2024 – Aug 2024

- Led the design and development of an automated stability and quality checking instrument set, focusing on ensuring precision and reliability in testing procedures.
- Programmed and integrated various laboratory instruments using multiple communication protocols, including VISA, serial, and USB, to facilitate seamless data acquisition and control.
- Managed the storage and organization of testing fibers, ensuring proper handling and inventory tracking.
- Collaborated with cross-functional teams to refine instrument calibration processes and optimize testing workflows, contributing to the overall efficiency and accuracy of the lab's research activities.

LEADERSHIP:

Corresponding Chair, *Institute of Electrical and Electronics Engineers - Student Branches*

Jan 2024 - Present

- Leveraged social media and digital platforms to effectively promote Electrical and Computer Engineering events, enhancing visibility and engagement within the student community.
- Demonstrated strong communication and marketing skills by creating and managing content that attracted participation and fostered community involvement.

Digital Technology Leader, *Mustang Marching Band*

Aug 2023 – Present

- Instruct and mentor students with no prior saxophone experience, fostering their musical development.
- Introduced new digital technology initiatives, including the design of custom-built accessories for band use.